WASP: Novel metal oxide nanomaterial photo-electrocatalyst of pollutant breakdown and bio-phytoremediation of wastewater

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JC25

1. Introduction

1.1 Threat of wastewater

Industrial:
- Dyes: Environmental damage
- Heavy metals: Toxic

Agricultural:
- Organic waste: Eutrophication
- Pesticides: Ecosystem damage

Domestic:
- Water-borne diseases
- Plastic: choke sewage systems

Mishandling can have **catastrophic** and **wide-reaching** consequences.

1.2 Current wastewater management

NEWater:
Ultrafiltration, Reverse Osmosis, UV disinfection

Challenges faced

Energy intensive

Conflicts between water and emissions[1]

Methane emissions from sludge waste

1.3 Aims

Aims: To develop a Wastewater Advanced Sustainable Pre-treatment (WASP) system which:

- Reduces energy intensity
- Relies on natural flora and fauna
- Functions as a self-sustaining mini ecosystem

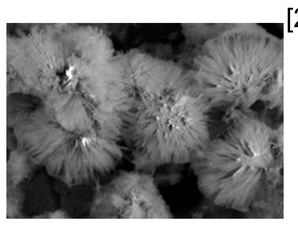
Pre-treated water can be further treated with **much less energy** needed.

2. Solution and Prototype

2.1 WASP pre-treatment stages

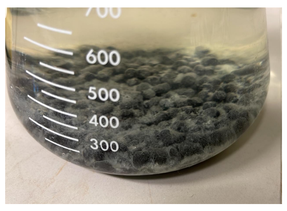
Stage 1. Photo-electrocatalysed dye breakdown

- Utilise light in **photovoltaic** applications to **break down dyes**
- Unique nano-shapes increase **porosity** and **rate of breakdown**
- E.g. Zinc Oxide nanoflowers[2]


[2]

Stage 2. Bio-remediation

- Bacteria can biosorp heavy metals, making them **less harmful**
- Biodegrade organic pollutants
- Biochar matrix (from sugarcane bagesse) helps **increase surface area**



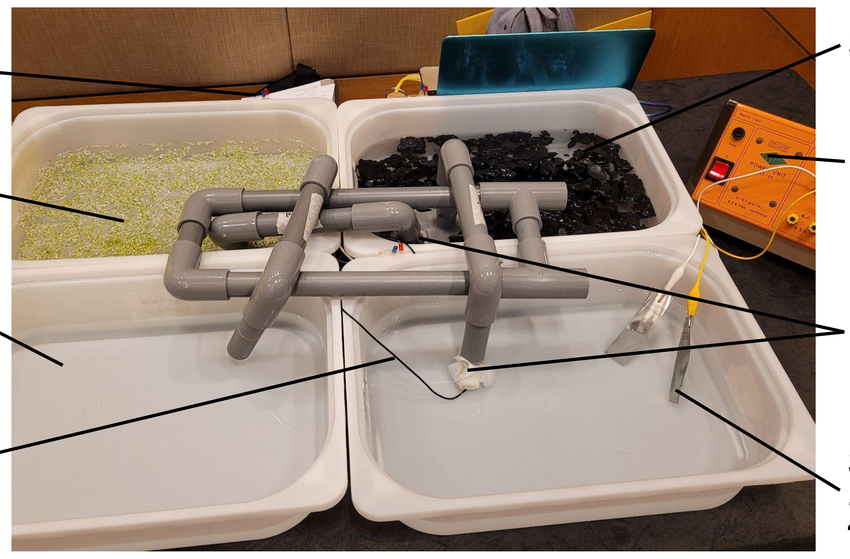
Stage 3. Phyto-remediation

- Can **remove** lead and cadmium
- Absorb organic waste for their own growth, **reducing carbon pollutants**
- Can even act as a **food source**

2.2 Flow of wastewater

Source of wastewater → Filtration for big pieces of debris → WASP multi-step pretreatment → **Lower-intensity** further treatment

2.3 Final Prototype



3. Application of STEM

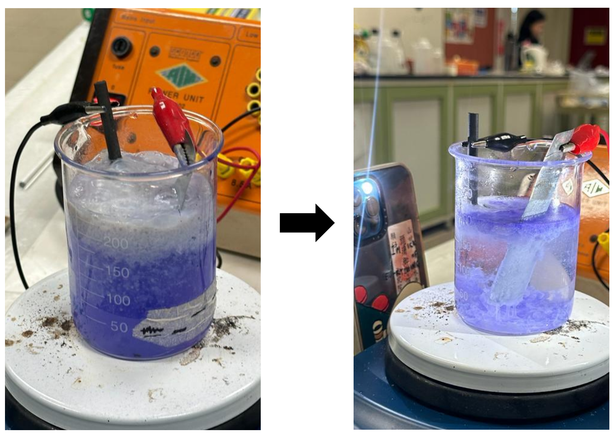
Science
- Experimentation of effectiveness of various stages
- Synthesis of ZnO nanoflowers

Engineering
- Context-specific design of overall prototype
- Design of pump system and electrical wirings

Technology
- Programmatic timed control of pump system and LEDs with Arduino

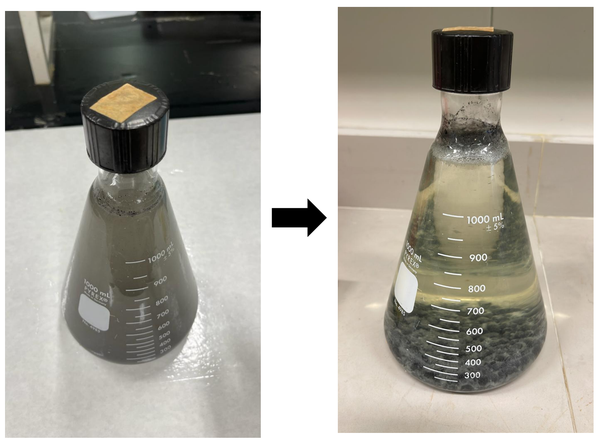
4. Results and Discussion

4.1 Stage 1



- Dye coagulates and is filtered off
- By spectrophotometry, concentration **reduced by 99%**

4.2 Stage 2



- Biochar provided excellent substrate, forming **biofilm**
- Contaminants **broken down** and **absorbed** by biofilm

4.3 Stage 3

- Duckweed absorbed excess nutrients and **multiplied rapidly**
- Suggests most harmful substances **removed** by stages 1 and 2

5. Conclusion

Findings

1. Stages are **effective** in removing dyes, heavy metals and organic waste
2. Prototype incorporates all elements and WASP system is **feasible**, **meaningful** and **scalable.**

Application

➔ In water treament plants before further treatment: less energy use

➔ Duckweed potentially used as a safe source of food

Future work

➔ Measure concentration of pollutants in incoming water or Singapore wastewater

➔ **Dynamically adjust** duration and intensity of each stage based on **concentration**

[1] Ming, S. (2016, April 23). The Big Read: Water security, energy use a difficult trade-off. Today Online. https://www.todayonline.com/singapore/water-security-energy-use-difficult-trade

[2] AlGhamdi, M. G. (2023). A multi-functional and highly efficient photo-electrocatalyst 3D ZnO nanoflowers for simultaneous water splitting and wastewater treatment. International Science and Engineering Fair 2023. https://projectboard.world/isef/project/enev046-3d-zno-for-wastewater-splitting-and-treatment

Images taken by contestant unles otherwise stated.